

Oxford Handbooks Online

Evolutionary Economic Geography

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The New Oxford Handbook of Economic Geography

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Print Publication Date: Jan 2018 Subject: Economics and Finance, Business Economics

Online Publication Date: Feb 2018 DOI: 10.1093/oxfordhb/9780198755609.013.11

Abstract and Keywords

The chapter gives a brief overview of the most recent, relevant literature on evolutionary economic geography. We describe how evolutionary economic geography has provided new and additional insights on a number of topics that belong to the core of the economic geography discipline: why do industries concentrate in space, how do clusters operate and evolve, how are innovation networks structured in space and how do they evolve over time, what types of agglomeration externalities induce urban and regional growth, how do regions diversify, and how do institutions and institutional change matter for the development of new growth paths in regions?

Keywords: evolutionary economic geography, related variety, regional branching, proximity, path dependence, co-evolution, institutional change

Introduction

FROM its start, evolutionary economic geography (EEG) has aimed to contribute to the understanding of several topics in economic geography, as to why industries concentrate in space, how networks evolve in space, and why some regions grow more than others. These core topics are by no means new in economic geography, but theorized and analysed from an evolutionary perspective, EEG provides new and additional insights, and in some cases alternative explanations (Boschma and Lambooy, 1999; Boschma and Frenken, 2006; Martin and Sunley, 2006; Boschma and Martin, 2010; Kogler, 2015; special issues on EEG in *Journal of Economic Geography* 2007, *Economic Geography* 2009, *Regional Studies* 2015, and *Journal of Economic and Social Geography* 2015). The main objective of this chapter is to provide a short outline of key contributions of EEG with respect to industrial clusters ('Clustering as an Evolutionary Process'), geography of

networks ('Networking as an Evolutionary Process'), urban and regional development ('Regional Development as an Evolutionary Process'), and the role of institutions ('Institutions and Evolutionary Economic Geography').

Evolutionary Economic Geography

Key to the emergence of EEG has been the establishment of evolutionary economics (Nelson and Winter, 1982; Dosi et al., 1988), which presented itself in the 1980s as an alternative to orthodox or mainstream neoclassical economics, rejecting the representative agent, optimal decision-making, and equilibrium analyses. Because the evolutionary critique on mainstream economics appealed to many economic geographers, while evolutionary economics itself was basically a-spatial (Boschma and Lambooy, 1999), economic geographers started (p. 214) to develop new concepts based on evolutionary thinking, including 'windows of locational opportunity' (Storper and Walker, 1989), 'technology districts' (Storper, 1992), and 'regional innovation systems' (Cooke, 1992).

This led to a lively debate about how to apply evolutionary principles to economic geography in a coherent manner. Several bodies of literature have served as key sources of inspiration to evolutionary scholars in economic geography in particular (Boschma and Martin, 2010): generalized Darwinism (Essletzbichler and Rigby, 2010), path dependency theory (Martin and Sunley, 2006), complexity theory (Martin and Sunley, 2007), and geographical political economy (MacKinnon et al., 2009). What these evolutionary approaches to economic geography have in common is a focus on historical processes that explain the uneven development and transformation of the economic landscape. This spatial pattern is perceived as the outcome of largely contingent, yet path-dependent and place-dependent historical processes.

Since the late 1990s, EEG slowly developed into an alternative to equilibrium-based perspectives dominant in the 'new economic geography' and in large parts of the regional science literature (Boschma and Frenken, 2006). At this time, most work in EEG built on biological notions as variety, selection, retention, mutation, and adaptation, as well as economic notions such as path dependence, lock-in, and proximity (Boschma and Frenken, 2003, 2006; Martin and Sunley, 2006). It often has a strong micro-foundation, taking organizational routines (Nelson and Winter, 1982) as a point of departure. Routines are capabilities on the basis of which organizations, like firms, try to adapt and survive in a dynamic environment. Economic evolution is understood as the creation and selective transmission of new routines in space. This transmission process is imperfect because of bounded rationality, as actors have no full access nor a perfect ability to respond to information, and because of spatial conditions (like institutions) that favour (or not) the creation and diffusion of routines. This leads to uneven geographical patterns, as embodied in agglomerations, centre-periphery patterns, clusters, and networks. The main focus of EEG is on how these spatial structures of the economy emerge from and are

transformed by the micro-behaviour of individual and collective agents, and why and how these processes of change are themselves path- and place-dependent (Boschma and Frenken, 2006).

Within this alternative framework to equilibrium-based perspectives, EEG has made a number of contributions to the field of economic geography. Firstly, EEG has challenged the Marshallian view on industry clustering in economic geography, in which firms in clusters would automatically benefit from Marshallian externalities, and clusters are conceived to exist and persist because of that. Secondly, EEG has proposed a dynamic proximity view on the geography of knowledge networks that has led to additional insights in the cluster and network literatures. Thirdly, EEG has made new contributions to the agglomeration externalities literature with the introduction of the concept of related variety, and it has initiated a new literature on regional diversification, known as regional branching. Fourthly, EEG has taken a critical view on the way institutions have been analysed in economic geography, by drawing attention, for example, to the contingent role of (local) institutions, as exemplified by the existence and persistence of heterogeneity of actors in the same institutional environment (Boschma and Frenken, 2009). In what comes next, we elaborate on each of these key contributions of EEG one by one.

(p. 215) **Clustering as an evolutionary process**

A classic question in economic geography is why some industries are concentrated in space. Marshall (1920) raised this question a century ago when he investigated the clustering of the metals industry in Sheffield and South Yorkshire in the UK (Potter and Watts, 2011). His explanation of such spatial clustering of an industry has dominated the field of economic geography for a long time: as soon as an industry locates somewhere, economic benefits can be derived from the co-location of firms in that industry, provided by a local pool of specialized knowledge, labour, and suppliers, also known as ‘localization economies’ or ‘Marshallian externalities’. What attracted special attention was Marshall’s remark that the ‘mysteries of trade’ in an industry were ‘in the air’ in clusters. His ideas implied that knowledge spillovers are geographically bounded, that intra-industry knowledge is accessible almost exclusively to firms in clusters, and that cluster firms are expected to perform better than firms outside clusters.

This dominant Marshallian thinking has been challenged, and EEG has made important contributions here. Sorenson and Audia (2000) and Stuart and Sorenson (2003) claimed that clusters emerge and exist because of a self-perpetuating process of local entry: the more local firms in a new industry, the more local entry. This has been called ‘cognitive legitimacy’ in organizational ecology, in which a high rate of local entry generates information that diffuses to potential entrepreneurs inducing them to create their own firms (Aldrich and Fiol, 1994; Maggioni, 2002; Wenting and Frenken, 2011). More local

entry also generates more local competition rendering it harder for cluster firms to survive. In this ecological view, clusters decrease entry costs, while increasing the chances of exit at the same time.

Klepper (2007) challenged Marshallian thinking even more by providing an alternative theory on spatial clustering from an industry life-cycle perspective. The point of departure is an evolutionary micro-perspective in which firms are depicted as being heterogeneous in their routines and capabilities because of bounded rationality, and therefore firms show differential growth rates. Firms have different routines because their pre-entry experience differs. In particular, spin-off companies inherit superior capabilities from successful parents from the same or related industries and therefore tend to outperform other types of entrants. As firms often locate in the founder's home region and rarely relocate (Stam, 2007; Dahl and Sorenson, 2012), a cluster can simply emerge through a local self-reinforcing spin-off process, and there is no need for Marshallian externalities to make that happen. Klepper (2007) found evidence for his spin-off thesis for the US automobile industry that concentrated in the Detroit region. He showed that survival rates depended primarily on the quality of pre-entry experience, not on Marshallian externalities (here, being located in Detroit). This finding has been replicated for other industries as diverse as semiconductors, tyres, fashion design, and book publishing. However, some studies, especially on creative industries, found that also Marshallian externalities, besides pre-entry experience, positively affected firm performance (for surveys, see Boschma, 2015a; Frenken et al., 2015).

Such an evolutionary approach explains not only why industries concentrate in space, but also why an industry historically emerged in one particular place, rather than another. As a cluster can, theoretically, emerge from a single successful firm generating many spin-offs, a cluster can emerge anywhere where this firm happens to locate (e.g. in the entrepreneur's (p. 216) home region). The question holds to what extent such a path-dependent process can be said to be also place-dependent (Martin and Sunley, 2006). Although the location of a cluster is rather unpredictable owing to the path-dependent, self-reinforcing logic of the spin-off process, there is substantial evidence that the first generation of successful firms in an emerging industry are often firms diversifying, or spinning off, from local related industries (Boschma and Wenting, 2007; Klepper, 2007; Buenstorf and Klepper, 2009). This is line with evidence that regions hosting industries that are related to a new industry have a higher probability of giving birth to it (Neffke et al., 2011).

Combined with Klepper's findings, recent studies suggest that clusters are characterized by positive related-industry externalities and (possibly) negative Marshallian externalities. In the conventional definitions of Marshallian externalities, as well as Porter's (1990) original cluster concept, intra-industry externalities and related-industry externalities are not analytically separated. This distinction, however, is very relevant because the two types of externalities tend to have opposite effects on firm survival. Related-industry externalities among local firms are expected to be positive, arising from knowledge spillovers and the mobility of skilled people, while intra-industry externalities

among local firms are expected to be mainly negative owing to competitive pressure and involuntary knowledge spillovers (Boschma, 2015a; Frenken et al., 2015). In particular, intra-industry externalities mostly harm well-performing cluster firms that have the most to lose and least to gain from other cluster firms, while young and small firms may still benefit from intra-industry externalities to compensate for their weak internal capabilities (Rigby and Brown, 2015). Similarly, firms are also heterogeneous to the extent to which they profit from the local presence of multinationals, with the most internationalized firms benefitting most (Crescenzi et al., 2015). Thus, given that firms are heterogeneous in their capabilities, the extent to which firms suffer or profit from co-location is expected to vary accordingly.

Another branch in the EEG literature is the 'cluster life-cycle' approach that studies the evolution of clusters, in particular the endogenous dynamics that may turn successful clusters into declining ones (Pouder and St. John, 1996; Brenner, 2004; Iammarino and McCann, 2006; Belussi and Sedita, 2009). It is crucial to underline that the life-cycle notion should be understood here in a non-deterministic, evolutionary manner (Martin and Sunley, 2011), as a cluster can renew itself, for instance. Menzel and Fornahl (2010) proposed a cluster life-cycle model in which firms enter and exit clusters, capabilities of firms interact and evolve, and inter-organizational linkages are formed and dissolved within and beyond clusters. When a cluster emerges, the heterogeneity of firms' capabilities initially increases but subsequently decreases, as firms engage in competition, inter-firm learning, and networking (Rigby and Essletzbichler, 1997; Vicente and Suire, 2007). If this convergence continues, the recombinant potential of the cluster decreases and its principal activities will decline. Menzel and Fornahl (2010) argue that a declining cluster can revive itself by upgrading its knowledge base through inflow of new knowledge from outside the cluster ('adaptation'), by integrating various local knowledge bases ('renewal'), or by diversifying into new activities while building on the local knowledge base ('transformation'). Clusters can only adapt if firms and other agents proactively engage in such a change process, but this is far from easy, given their proximity to local networks and institutions (Glasmeier, 1991). This 'lock-in' may be reinforced when public policy is responsive primarily to demands from vested interests (Grabher, 1993; Hassink, 2005), and local actors hold on to a collective identity (Staber and Sautter, 2011).

(p. 217) Life-cycle dynamics may also stem from herding behaviour in location decisions, indicative of hypes. Suire and Vicente (2009) developed a model of cluster emergence and stability that takes into account such herding effects. Firms may locate in clusters not for alleged Marshallian externalities associated with co-location, but for reasons of what Appold (2005) called 'geographical charisma'. Some clusters have a strong reputation owing to very visible and successful firms that attract other firms to the cluster. Here, being located in a cluster with successful firms acts as a signal to their stakeholders that they are present 'where the action takes place', hereby legitimating the location choice. The model by Suire and Vicente (2009) shows that if legitimation effects prevail, a cluster

can grow very fast, but remains fragile, as the pattern of co-location is not based on positive externalities. As a result, once the reputed firm would lose its reputation, or would relocate to another location, the cluster is likely to break down.

To conclude, the main contribution of EEG to the topic of spatial clustering of industries is that the dominant explanation of industry clustering resulting from Marshallian externalities has been challenged: (i) clusters can emerge despite the absence of localization economies; (ii) clusters can emerge and exist because of a self-reinforcing process of local entry, in particular the entry of successful spin-offs; (iii) emerging clusters tend to be characterized by positive related-industry externalities; (iv) not all firms perform equally in clusters—some have better routines, partly owing to the pre-entry background of the entrepreneur, and firms differ in their ability to exploit positive externalities and cope with negative externalities in clusters; (v) emergent clusters produce new institutions or adapt existing institutions by the collective action of agents; (vi) declining clusters can revive and overcome lock-in, but not necessarily so.

Networking as an Evolutionary Process

In economic geography, it is well known that spatial clustering provides opportunities to make connections between people and organizations. Firms not only compete, but also interact and collaborate with a range of organizations like other firms, banks, research institutes, or universities. As geographical distance often forms a barrier, organizations in the same region are more likely to connect, but not necessarily all of them. EEG is well equipped to incorporate these relational issues because it reasons from the heterogeneity of agents (Boschma and Frenken, 2010). As capabilities differ between organizations, they do not easily connect, nor do they easily learn from each other. This is exactly why networks in general, and knowledge and innovation networks in particular, are not randomly structured but skewed: that is, some organizations are more connected than others (Powell et al., 1996; Giuliani, 2007).

Such an evolutionary take on the geography of knowledge networks has led to additional insights in the cluster literature. Contrary to what the literature often suggests, being part of a cluster does not necessarily mean that all cluster firms are connected with each other. Indeed, there is strong evidence that some cluster firms are highly connected in (local) knowledge networks, while other cluster firms are poorly connected, or not connected at all. Giuliani and Bell (2005) showed in a seminal study on a Chilean wine cluster that firms with a high absorptive capacity occupy a more central position in the local knowledge network. (p. 218) Such firms are attractive partners to connect to and capable of absorbing knowledge from other firms in and outside the cluster. This means that only a few firms in clusters (the most connected) have access to crucial local knowledge. This goes against the Marshallian view that knowledge is ‘in the air’ in clusters, in which all cluster firms are perceived to have equal access to local knowledge

because they share the same location and the same norms and values. As Giuliani (2007) put it, knowledge networks are not pervasive but selective, and networks in clusters are no exception to that rule.

Besides individual features of firms, like absorptive capacity, proximities are also key drivers of network tie formation, and this is where the proximity literature and EEG clearly meet (Boschma and Frenken, 2010). As actors differ, they show a strong bias towards which firms they interact and collaborate with, preferring those with whom they share similar knowledge (cognitive proximity), norms and values (institutional proximity), the same location (geographical proximity), social ties (social proximity), or organizational boundaries (organizational proximity) (Boschma, 2005; Breschi and Lissoni, 2009). As other forms of proximity can substitute for geographical proximity, the proximity concept can explain why networks within clusters are not pervasive, and why some cluster firms, sometimes acting as gatekeepers (Morrison, 2008), have most of their relations with firms outside the cluster.

The relationship between proximity and firm performance is more ambiguous. While proximity promotes actors to collaborate, it does not necessarily increase collaborative performance, and may even turn out to be harmful. This has been referred to as the 'proximity paradox' (Broekel and Boschma, 2012). For instance, cognitive proximity not only facilitates communication and knowledge transfer between firms, but it also reduces the scope for learning and enhances the risk of involuntary knowledge leakage. Moreover, one expects proximity in relationships to increase over time, as interacting agents tend to become more similar as a result of social interaction and interactive learning (Balland et al., 2015). This has led to a search for 'optimal' proximity to cope with the negative aspects of proximity (Boschma, 2005). For geographical proximity, scholars have underlined the importance of a combination of local and non-local knowledge linkages for the long-term development of clusters (Asheim and Isaksen, 2002; Bathelt et al., 2004), or the importance of temporary proximity between agents who meet at conferences or trade fairs where they exchange knowledge (Torre, 2008). A questionable assumption in most proximity studies, however, holds the assumption of symmetry. A future challenge is to take up and integrate power and asymmetric relations in the proximity framework, as an actor can be proximate to another actor but not necessarily vice versa.

The evolution of networks has been a subject of recent research in EEG (Ter Wal and Boschma, 2011). Balland et al. (2013) studied network dynamics in the global video game industry by looking at collaborations of co-developers of new video games. Their study demonstrated that geographical proximity became a more important driver of network tie formation as the industry evolved. This increasing tendency of inter-firm collaboration at shorter geographical distances could be explained by the increasing technological complexity of video games (Sorenson et al., 2006) and the project-based nature of video game production in which 'local buzz' and 'who-knows-who' are key inputs (Grabher, 2006). Ter Wal (2014) found the opposite result in biotech: geographical proximity became less important as driver of co-inventor networks, possibly owing to the increasing codification of biotech knowledge. However, there still is little understanding of how

spatial networks change: little is known of how proximities in networks evolve over time (Balland et al., 2015), how network (p. 219) structures in clusters change, to what extent network dynamics exhibit path dependence (Gluckler, 2007), and how network dynamics affect the evolution of a cluster (Cantner and Graf, 2006; Hendry and Brown, 2006; Balland, 2012).

In sum, the contributions of EEG to the topic of spatial knowledge networks are, so far, the following: (i) knowledge is not 'in the air', but channeled through networks that are uneven and selective in clusters; (ii) networks are selective, because firms and other agents have different capabilities and routines; (iii) various proximities, including geographical proximity, are important drivers of network formation, but proximities do not necessarily increase the performance of firms; (iv) while geographical and institutional proximity may drive network tie formation in clusters, not all cluster firms will connect and perform equally, despite being part of the same local institutional environment; (v) network relations in clusters have a tendency to become more inward-looking over time; (vi) non-local linkages, or temporary proximities, are crucial for the competitiveness of cluster firms, but they require other forms of proximities to enable effective transmission of knowledge.

Regional Development as an Evolutionary Process

EEG has devoted attention to how regions can secure their long-term development by developing new industries or new growth paths. A source of inspiration has been Schumpeter's (1912) description of innovations as new combinations. This has been further developed in the notion of recombinant innovations, which emerge from recombining parts of pre-existing technologies or services in new ways (Fleming, 2001). When recombinant innovations are the rule rather than the exception, this implies that the existing variety in a region conditions the scope for innovation. This builds on the seminal work by Jacobs (1969, p. 59), who argued that 'the greater the sheer numbers and varieties of divisions of labor already achieved in an economy, the greater the economy's inherent capacity for adding still more kinds of goods and services. Also the possibilities increase for combining the existing divisions of labor in new ways'. This idea was taken up by Glaeser et al. (1992), who tested whether diversified or specialized regions tend to grow more. Diversified regions should be more innovative owing to Jacobs' externalities, while specialized regions could benefit from Marshallian externalities. Glaeser's study was followed by many others, but, despite massive empirical efforts, there is conflicting evidence for both hypotheses: there are almost as many studies proving that regions benefit from variety as there are studies showing that regions benefit from specialization (De Groot et al., 2009, 2015).

A possible reason for the weak evidence on Jacobs' externalities is that many technologies and services cannot be meaningfully combined. Rather, one expects that recombinant innovations more often stem from related industries that share similar knowledge and skills. Frenken et al. (2007, p. 687) argued that for variety to be supportive in innovation processes, variety must be related (i.e. cognitively close), as related variety 'improves the opportunities to interact, copy, modify, and recombine ideas, practices and technologies across industries giving rise to Jacobs externalities'. This motivated studies to test whether related variety increases regional employment growth. The evidence collected so far indicates by and large a (p. 220) positive effect of related variety on employment growth (Essletzbichler, 2007; Frenken et al., 2007; Quatraro, 2010), especially in knowledge-intensive industries (Bishop and Gripiaios, 2010; Hartog et al., 2012).

The question of new industry formation is associated with the concept of related variety. Frenken and Boschma (2007) depicted local industry formation as a branching process in which the local presence of industries that are related to a new industry increases the probability for a new industry to occur, given that related industries provide the main source for knowledge, capabilities, and potential entrepreneurs (Klepper, 2007). The more related the variety of industries is vis-à-vis the new industry, the more likely a region can be successful in that new industry. Hence, the existing set of industries conditions the likelihood of new industries emerging, and in that sense there exists 'regional path dependence' (Iammarino, 2005; Martin and Sunley, 2006; Fornahl and Guenther, 2010).

Empirically, the branching phenomenon has been analysed at the level of countries by Hidalgo et al. (2007), who demonstrated that countries tend to develop new export products that are related in 'product space' with existing export products. The product space specifies the relatedness between products based on the frequency of co-occurrence of products in countries' export portfolios. The same reasoning has been applied to understand the development of regions becoming active in new markets. Neffke et al. (2011) found that an industry had a higher probability of entering a region when technologically related to pre-existing industries in that region. Studies have confirmed relatedness driving regional diversification in new industries (Boschma et al., 2013; Essletzbichler, 2015), new technologies (Kogler et al., 2013; Rigby, 2015), and new eco-technologies (Tanner, 2014; van den Berge and Weterings, 2014).

What these studies tend to show is that related diversification in regions is the rule and unrelated diversification the exception. That unrelated diversification is a more rare event does not come as a surprise, as it is more uncertain and risky to recombine previously unrelated domains. It is a crucial question whether regions can keep relying on related diversification to sustain long-term development, or whether they need to diversify in unrelated activities now and then. Studies have started investigating the conditions that make regions more likely to diversify into unrelated activities. Castaldi et al. (2015) found that unrelated variety is associated with high rates of breakthrough innovations in US states. In the rare cases that recombination innovations between unrelated technologies or services succeed, they become related (Desrochers and

Leppälä, 2011). Such a radical new combination not only opens up new markets and innovation opportunities, but it might also provide the basis for long-lasting competitive advantage, as other regions will face difficulties in coping with such radical change. A similar issue is analysed in the expanding literature on new growth paths (Garud et al., 2010) in which new path creation is defined as the emergence of entirely new sectors or products, while path renewal occurs when local activities switch to different but related activities (Isaksen and Trippl, 2014). To break with path dependence and create new growth paths, regions will have to rely more on knowledge and resources residing in other regions. Hence, the presence of multinationals, the immigration of entrepreneurs, and a targeted government policy are all elements that come into play in explaining new path creation (Binz et al., 2013; Dawley, 2014; Neffke et al., 2017).

The contributions of EEG on regional development so far are: (i) related variety is a key concept in EEG that has shed new light on the MAR (Marshall–Arrow–Romer) versus Jacobs’ externalities debate—there is emerging evidence of positive externalities stemming (p. 221) from the co-presence of firms in related industries; (ii) EEG has shed light on how regions diversify over time—regional development is depicted as a branching phenomenon in which new recombinations stem from related activities that share similar knowledge and skills, and in which local capabilities in existing industries or technologies conditions the set of industries and technologies that are more likely to emerge; (iii) unrelated diversification, recombining previously unrelated fields, is expected to be a more rare event, and tends to rely more on the inflow of resources and capabilities from other regions.

Institutions and Evolutionary Economic Geography

A recurrent critique to evolutionary scholars in economic geography has been the perceived neglect of the role of institutions in firm behaviour and economic development processes (MacKinnon et al., 2009). This critique is understandable given that many empirical studies in EEG did not pay explicit attention to the institutional contexts in which economic processes take place, or ‘bracketed’ such processes in dummy variables or ‘fixed effects’.

However, this relative empirical neglect says little about the theoretical possibilities to integrate institutional analysis into the EEG framework. EEG has engaged at length with the question of how institutional and evolutionary approaches can be combined (see e.g. Boschma and Lambooy, 1999; Boschma, 2004; Boschma and Frenken, 2009; Martin, 2010). Institutions provide incentives, but they may also form obstacles that make the development of some industries and organizational practices in some places more feasible (Malmberg and Maskell, 2010). Institutions are depicted as co-evolving with new technologies and markets that are deemed crucial for the development of new industries

(Nelson, 1994). The chances for new industry formation in a region depend on the timing and direction in which institutions are adapted in a way that supports the industry's further development (Murmman, 2003). This requires more understanding of the conditions that favour or hamper institutional change in regions. In an attempt to come to a theory of institutional change and new industry formation, Battilana et al. (2009) argued that conditions supportive of institutional change are a common sense of urgency (e.g. due to a crisis), institutional contradictions and discontent (e.g. as new industries challenge existing categorizations), and a low degree of past institutionalization. These conditions may have a strong regional dimension, suggesting that regions are not all equally likely to engage in effective institutional change.

EEG has also made progress in taking up explicitly the role of institutions in recent empirical work. There is an increasing attention paid to how local agents (private and public) engage in collective action to mobilize knowledge, resources, and public opinion as to create new or adapt existing institutions, how vested interests may be circumvented, and the key role that both regional and national governments can play in regional economic development (Feldman et al., 2005; Strambach, 2010; Sotarauta and Pulkkinen, 2011; Binz et al., 2013). In quantitative studies, the role of institutions has been highlighted as well. In their study on the local entry dynamics of fashion designers across the world, Wenting and Frenken (2011) could show that the institutional environment in Paris blocked the starting (p. 222) up of new firms in commercial design owing to strict regulations in Paris's design profession, while designers in other cities did not experience such obstacles. A recent study by Boschma and Capone (2015), building on the literature on varieties of capitalism (Hall and Soskice, 2001), found that institutions associated with 'liberal market economies' give countries considerably more freedom to diversify in more unrelated activities than institutions associated with 'coordinated market economies', thus shedding light on different development logics channelled by national institutional environments.

An issue that has remained little elaborated is how the concept of innovation system can be integrated in EEG. The regional innovation system (RIS) literature has explained the clustering of innovative activities by focusing on the nature of relationships between organizations such as firms, governments, universities, and non-governmental organizations that are involved in the innovation process at the sub-national level (Cooke, 1992; Asheim and Isaksen, 1997). Having strong evolutionary roots (Freeman, 1987), this approach has drawn attention to the importance of localized capabilities for the production and transmission of tacit knowledge (Asheim and Gertler, 2005). Given the path-dependent nature of building up localized capabilities, as embodied in local knowledge bases and institutions, it is hard for regions to imitate 'constructed regional advantages' from successful regions (Asheim et al., 2011). Recently, scholars have expressed a need to go beyond a static approach that maps actors and institutions in a RIS, and to concentrate more on how RIS change in response to globalization, technological change, and societal challenges. This necessitates an understanding of how changes in RIS are initiated and implemented by agents, how changes in institutions are activated, and how relations at multiple spatial scales are constructed, managed, and

utilized. Recently, there is increasing attention to the global dynamics that underlie the early formation of new innovation systems (Binz et al., 2013, 2014). The understanding of where new technologies emerge requires not only insight into the local mechanisms of capability transfer from related technologies to emerging ones, but also into the organization of knowledge production and regulatory processes at the national and international level (Morrison and Cusmano, 2015).

EEG has a particular take on the role of institutions: (i) the influence of (local) institutions is contingent given the existence and persistence of heterogeneity of firms in the same institutional context; (ii) institutions have an effect on the intensity and nature of interactions between agents in RIS, and therefore they affect the process of regional diversification; (iii) new industry formation is depicted as co-evolving with the establishment of new institutions or the adaptation of existing ones; (iv) local agents engage in collective action to create new or adapt existing institutions and challenge vested interests that may oppose such change; (v) regions may differ in their ability to induce required institutional change; (vi) there is still little understanding of what conditions at various spatial scales support or hamper institutional change.

Final Remarks

This chapter has given a brief outline of recent theoretical and empirical contributions of EEG with respect to clustering, networking, urban and regional development, and the role (p. 223) of institutions. Give the limited space available, this outline has been partial at best. Studies in EEG have provided new but often still preliminary answers to old enduring questions in economic geography, and also bring up new questions and problems not yet explored. Although advancing, the empirical literature on EEG is work in progress. This also applies to the development of its main concepts like institutions (MacKinnon et al., 2009), path dependence (Martin, 2010), life cycle (Martin and Sunley, 2011), development (Martin and Sunley, 2015), and the use of appropriate methodologies that do justice to evolutionary principles (Hassink et al., 2014).

Recently, EEG has been moving into topics like regional resilience (Simmie and Martin, 2010; Boschma, 2015b), the geography of transition (Truffer and Coenen, 2012; Patchell and Hayter, 2013), and public policy and governance, especially in the context of the smart specialization debate (Foray, 2015; McCann and Ortega-Argiles, 2015). An EEG approach on resilience aims to leave behind an equilibrium perspective that is primarily interested in a recovery to pre-existing or new equilibrium states, and instead proposes an evolutionary perspective in which regional resilience is redefined and analysed in terms of the impact of shocks on the capacity of regions to develop new growth paths (Boschma, 2015b). The newly emerging literature of the geography of transition (Truffer and Coenen 2012) is also promising, as it provides a comprehensive view on niche formation and the role of collective agents that has not yet been fully developed in EEG. In turn, EEG provides concepts like regional branching that describe how regions move into new green technologies, and why some regions are more successful in doing so (Tanner, 2014; van den Berge and Weterings, 2014). The smart specialization literature explores how the entrepreneurial discovery process can be organized and governed to make regions move in new specializations. To the main adherents of smart specialization strategies, related variety and regional branching, are regarded as key building blocks of smart specialization policies (Foray, 2015; McCann and Ortega-Argiles, 2015). Without doubt, this future work on smart specialization will boost the further development of regional innovation policy in an EEG framework that is far from fully developed (Coenen et al., 2015).

These continuing debates and emerging topics bear testimony to the fact that EEG is alive and evolving, and they will surely make EEG develop further into a more comprehensive approach in economic geography.

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